ESE_ December 2020_Computer_ComputerNetworks_ 2UCC502_ 5th_ 26/12/2020_ 2:30PM_ Objective Questions (20 Marks)

* Required

1. Email address *

2. Enter your Full Name *

3. Enter Your Roll no. *

4. Enter your division *

Mark only one oval.

- A
- B

Questions

5. Consider the following statements:- A.) TCP connections are full duplex B.) TCP has no option for selective acknowledgement C.) TCP connections are message streams

1 point

Mark only one oval.

- Only A is correct
- Only A and C are correct
- Only B and C are correct
- All of A, B, and C are correct

6. In one of the pairs of protocols given below, both the protocols can use multiple TCP connections between the same client and the server. Which one is that?

1 point

Mark only one oval.

- FTP, SMTP
- HTTP, TELNET
- HTTP, FTP
- HTTP, SMTP

7. Which one of the following fields of an IP header is NOT modified by a typical IP router?

1 point

Mark only one oval.

- Checksum
- Source address
- Time to Live (TTL)
- Length

8. A bit-stuffing based framing protocol uses an 8-bit delimiter pattern of 01111110. If the output bit string after stuffing is 01111100101, then the input bit-string is 1 point

Mark only one oval.

- 0111110100
- 0111110101
- 0111111101
- 0111111111

9. In the following pairs of OSI protocol layer/sublayer and its functionality, the INCORRECT pair is 1 point

Mark only one oval.

- Network layer and Routing
- Data Link Layer and Bit synchronisation
- Transport layer and End-to-end process communication
- Medium Access Control sub-layer and Channel sharing

10. Which one of the following is TRUE about the interior gateway routing protocols–Routing Information Protocol (RIP) and Open Shortest Path First (OSPF)? 1 point

Mark only one oval.

- RIP uses distance vector routing and OSPF uses link state routing
- OSPF uses distance vector routing and RIP uses link state routing
- Both RIP and OSPF use link state routing
- Both RIP and OSPF use distance vector routing

11. If the total length bits are 0000000000111111 and HLEN = 1010, then which one is true about the size of header and payload?

1 point

Mark only one oval.

- 40 and 63
- 40 and 23
- 24 and 63
- 24 and 23

12. CRC can detect all burst error of up to M errors, if generator polynomial G(x) is of degree

1 point

Mark only one oval.

- It does not matter
- M − 1
- M/2
- M + 1

13. In Ethernet CSMA/CD, the special bit sequence transmitted by media access management for collision handling is called

1 point

Mark only one oval.

- Hamming code
- CRC
- Jam
- Preamble

14. Which of the following functionality must be implemented by transport layer over and above the network protocol? 1 point

Mark only one oval.

- Recovery from packet loss
- End to end connectivity
- Packet delivery in encapsulated abstract order
- Secure transmission from end to end

15. If 5-bit sequence number is used, what is the sequence number after sending 100 frames in Go-Back-N ARQ? 2 points

Mark only one oval.

- 2
- 3
- 4
- 5

16. Host A sends an IP datagram D to host B. Assume that no error occurred during the transmission of D. When D reaches B, which of the following IP header field(s) may be different from that of the original datagram D? (i) TTL (ii) Checksum (iii) Fragment Offset 2 points

Mark only one oval.

- (i) only
- (i) and (ii) only
- (ii) and (iii) only
- (i), (ii) and (iii)

17. An IP router with a Maximum Transmission Unit (MTU) of 1500 bytes has received an IP packet of size 4404 bytes with an IP header of length 20 bytes. The values of the relevant fields in the header of the third IP fragment generated by the router for this packet are

2 points

Mark only one oval.

- MF bit: 0, Datagram Length: 1444
- MF bit: 1, Datagram Length: 1424
- MF bit: 1, Datagram Length: 1500
- MF bit: 0, Datagram Length: 1424

18. If N is the maximum sequence number, then the window size in selective repeat and Go-Back-N protocols are, respectively,

2 points

Mark only one oval.

- N/2, N − 1
- N/2, N + 1
- N + 1/2, N − 1/2
- N − 1, N/2

19. Two computers C1 and C2 are configured as follows. C1 has IP address 203.197.2.53 and netmask 255.255.128.0. C2 has IP address 203.197.75.201 and netmask 255.255.192.0. Which one of the following statements is true?

2 points

Mark only one oval.

- C1 and C2 both assume they are on the same network.
- C2 assumes C1 is on the same network, but C1 assumes C2 is on a different network.
- C1 assumes C2 is on the same network, but C2 assumes C1 is on a different network.
- C1 and C2 both assume they are on different networks.